

Simulating breeding programs: Pigs

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Overview

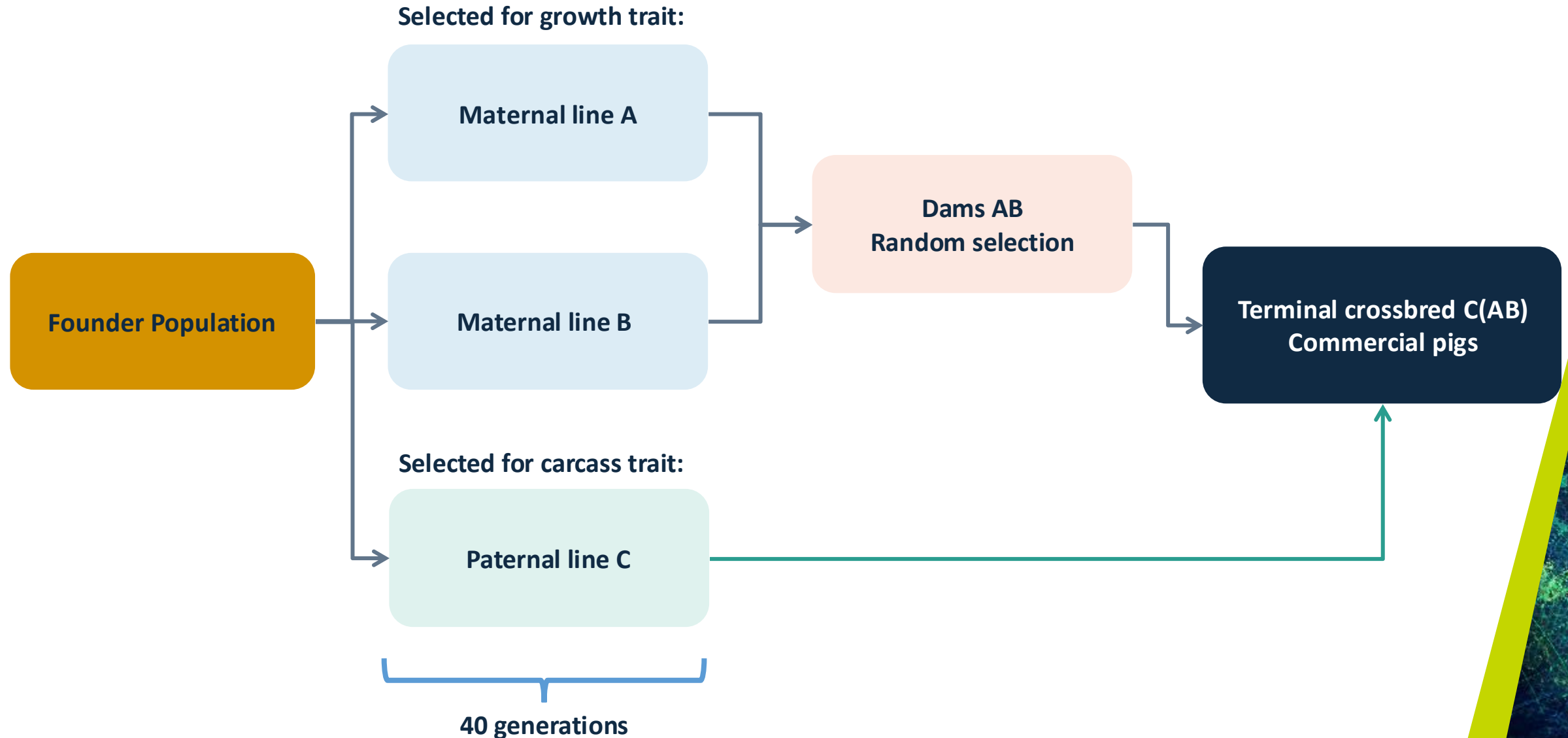
Simulating a 3-way crossbreeding system

1. Founder population & pure lines
2. Trait definition & genetic architecture
3. 40 generations of selection
4. Crossbreeding & terminal offspring
5. Phenotypic response & stochastic variation

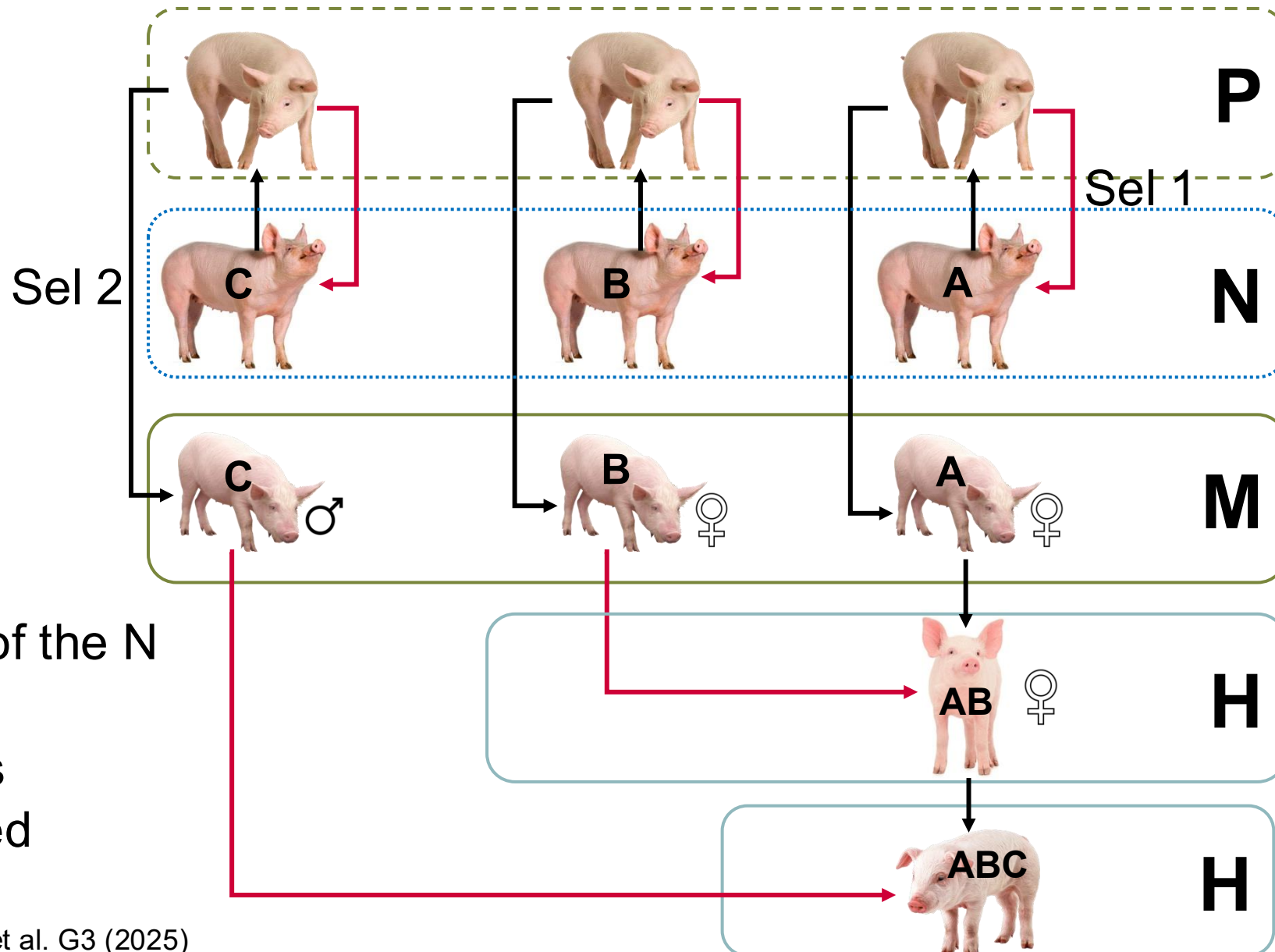
Three-way crossbreeding system



Breeding program structure

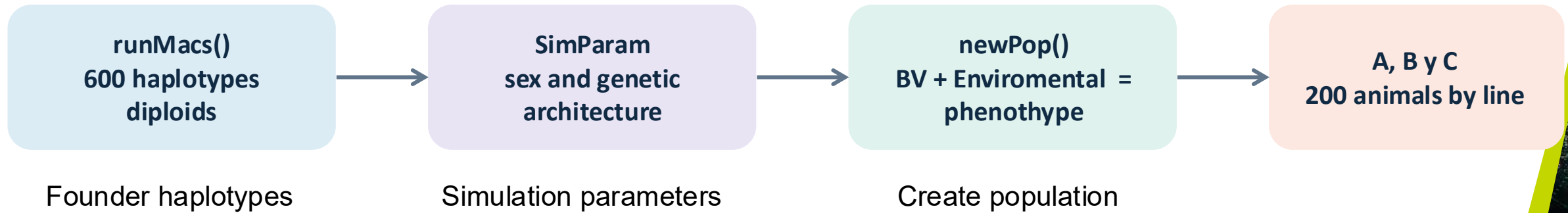


A classical 3-way crossbreeding system with multipliers for different breeds



P: Offspring of the N
N: Nucleus
M: Multipliers
H: Crossbreed

From haplotypes to phenotype



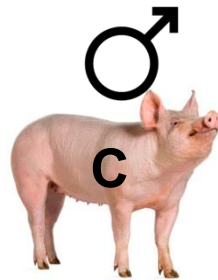
The initial separation creates genetic replicas, not already divergent historical breeds.

Phenotypic traits

Quantitative traits

$$y \sim N(\mu_y, \sigma_y^2)$$

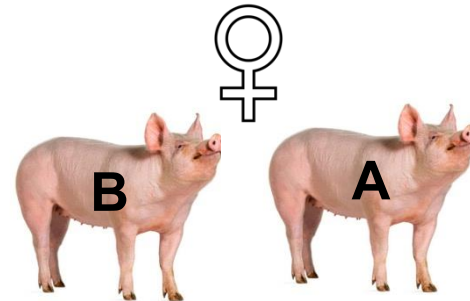
Ribeye/Loin eye area



$$h^2 = 0.30$$

$$LEA \sim N(45, 9)$$

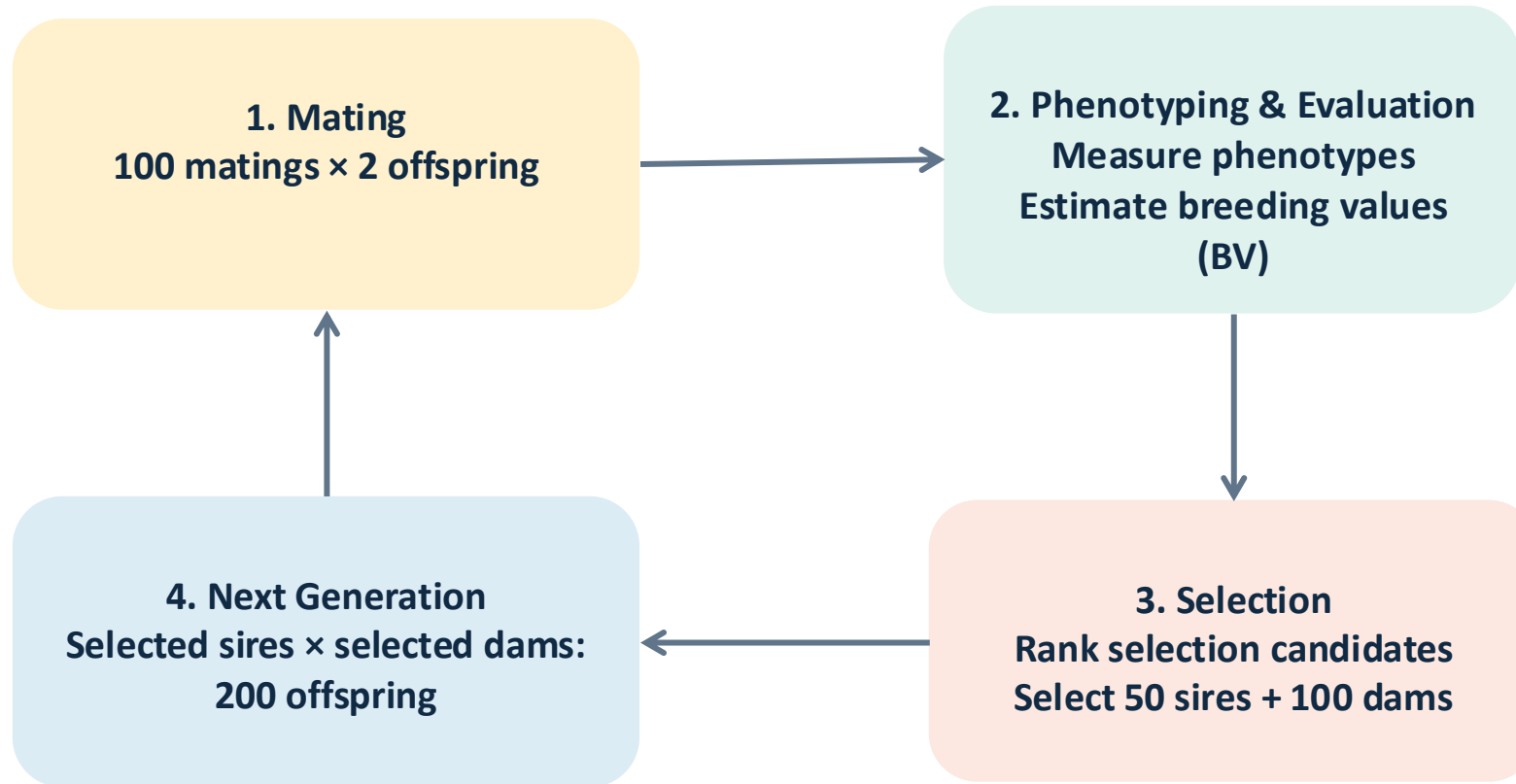
Average daily gain



$$h^2 = 0.40$$

$$ADG \sim N(0.8, 0.01)$$

Within-line selection cycle



A y B: trait = 1 (ADG) C: trait = 2 (LEA)

Crossbreed population

Line A: 100 Dams

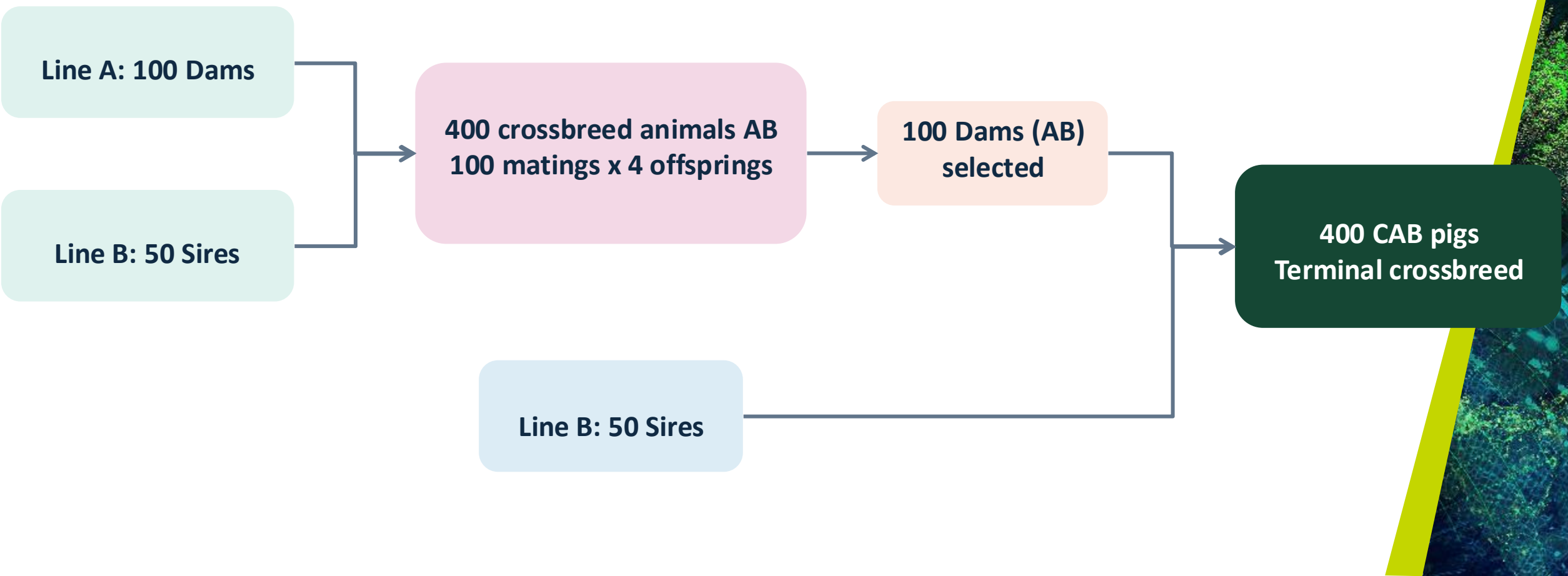
Line B: 50 Sires

400 crossbreed animals AB
100 matings x 4 offsprings

100 Dams (AB)
selected

Line B: 50 Sires

400 CAB pigs
Terminal crossbreed



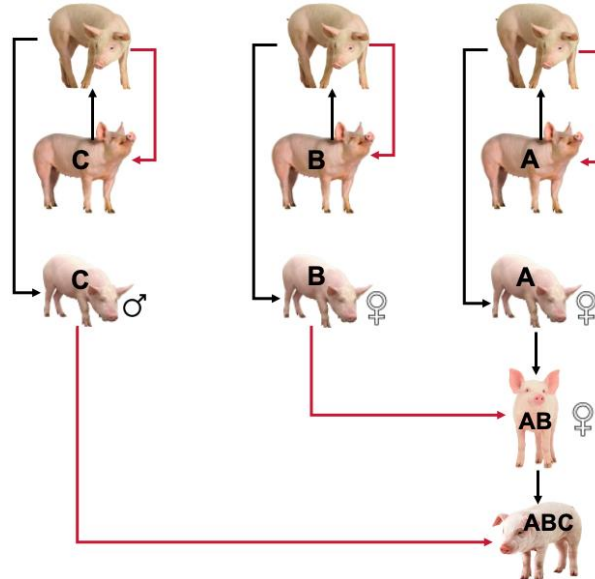
Applications of simulation in pigs breeding

How can commercial crossbred performance be improved using information from purebred and crossbred animals?

1. Inclusion of crossbred information (See et al., 2020)
2. Which crossbred animals should be genotyped? (See et al., 2021)
3. Imputation of non-genotyped F1 females (See et al., 2022)

1. Inclusion of crossbred information

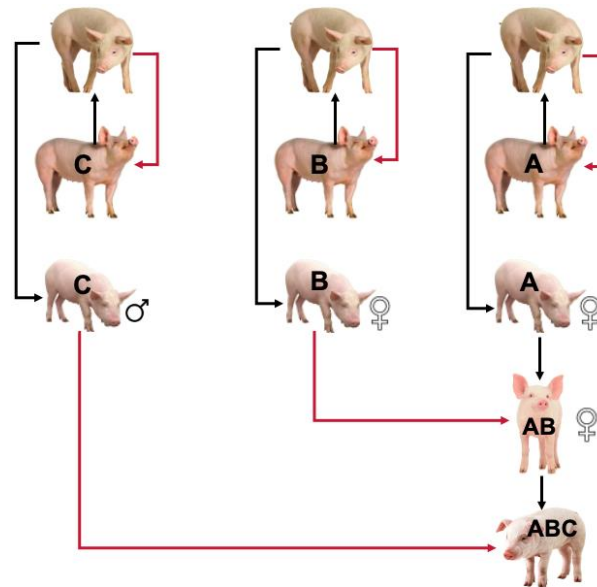
- **Objective:** To determine how much genetic selection improves when phenotypes and genotypes from crossbred animals are included in the genetic evaluation of purebred lines.



Different genetic correlations between purebred and crossbred performance: $r_{pc} = 0.1, 0.3, 0.5, 0.7, 0.9$

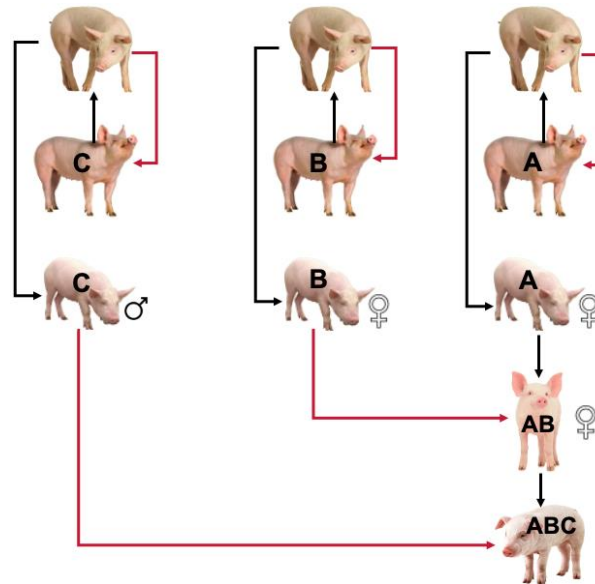
2. Which crossbred animals should be genotyped?

Objective: To identify the most efficient strategy for selecting which crossbred animals to genotype when there is not enough funding to genotype all of them.



3. Imputation of non-genotyped F1 females

Objective: To evaluate whether the genotypes of F1 females could be reconstructed without directly genotyping them and whether this information could improve selection.





AlphaSimR manual

<https://gaynorr.r-universe.dev/AlphaSimR/doc/manual.html>



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